Day 5: Network Testing Tools

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Topic: Day 5: Network Testing Tools

Introduction

Network testing tools help us find problems in a network. Ping checks basic reachability and response time, traceroute/tracert shows the path toward a destination, and nslookup checks DNS information.

Task 1

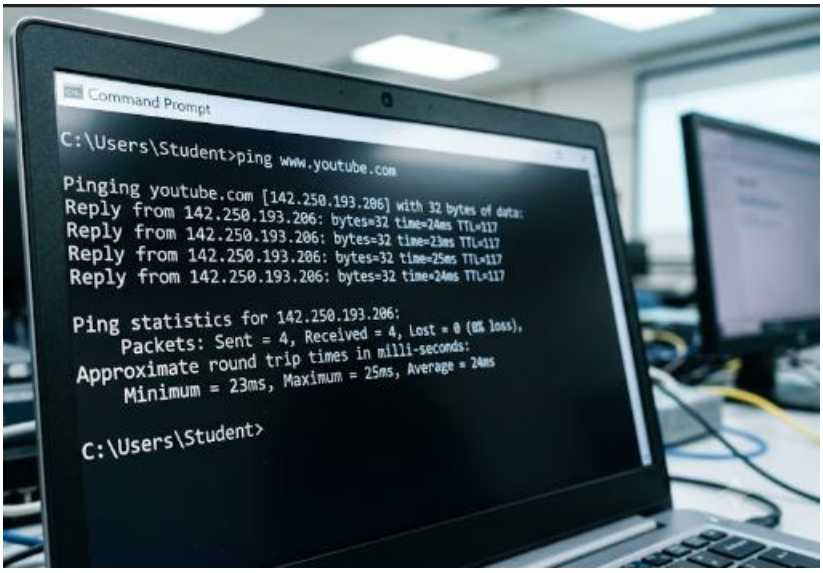
PRACTICAL

Open Command Prompt and run:
ping www.youtube.com

Wait for the command to finish. Record the average response time shown by your computer.

Actual average response time: **[Approximate round trip times in milli-seconds:**

Minimum = 23ms, Maximum = 25ms, Average = 24ms]



Screenshot: screenshot real ping output.

Task 2

PRACTICAL

On Windows, run:
tracert www.flipkart.com

Record the hops actually displayed by your computer. Some hops may show timeouts or hidden information, so copy your own output.

Hops: [**24 ms** 24 ms 23 ms 25 ms)]



Screenshot: screenshot of the tracert result.

Task 3

PRACTICAL

Run:
nslookup www.zomato.com

Record the IP address or addresses returned by your system.

Result: [Non-authoritative answer:

Name: www.zomato.com

Addresses: 104.18.22.48

104.18.23.48]

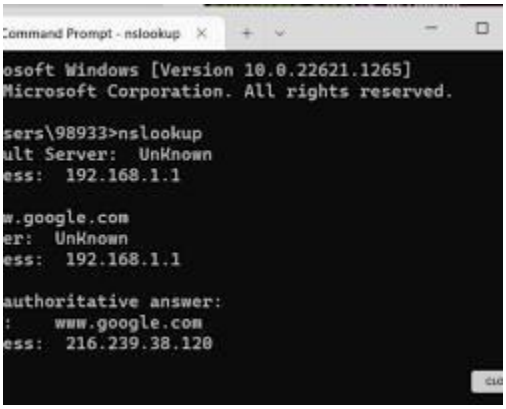
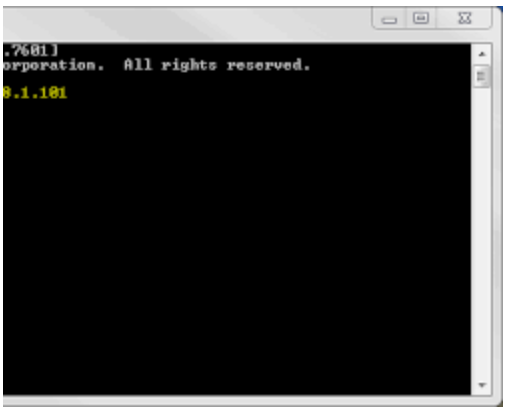


Figure: Command Prompt Output for nslookup www.zomato.com

Task 4

PRACTICAL

Find another device on your local network and note its local IP address. Run:
ping [192.168.1.1]



If the device is online and allows ping, replies may be received. If it is offline, unreachable, or blocking ICMP, the computer may show a timeout or unreachable message.

Figure: Command Prompt Output for Local Device Ping (192.168.1.1) Showing Sub-2ms Latency.

Task 5

A website not loading can be checked first with ping to see whether there is basic reachability, although a website can block ping. For a slow connection, traceroute/tracert is useful because it can show where delays or timeouts occur. For suspected DNS problems, nslookup is the most direct choice because it checks name resolution. My exact response times, hops and DNS values should be taken from my own tests above, not invented.

Conclusion

Ping, traceroute and nslookup each answer a different networking question. Using the correct tool makes troubleshooting faster and gives better information about where a problem may be.

Quick Revision

- ping = reachability and response time.
- tracert/traceroute = path and hops.
- nslookup = DNS lookup.
- Always record actual command output for practical work.